# Tutorial: Neutron Scattering Lengths

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# Tutorial: Neutron Scattering Lengths

- In MDANSE, we use the values of `b_coherent` and `b_incoherent` (neutron scattering lengths) to determine the contribution of atoms to the total neutron scattering result.
- Let's consider hydrogen: H1, H2 and H3 are labels for specific hydrogen isotopes: $^{1}H$, $^{2}H$ and $^{3}H$.
- The fourth one, H, is an averaged atom type with scattering lengths which depend on the natural abundances of the three isotopes.

https://www.ncnr.nist.gov/resources/n-lengths/

| Neutron scattering lengths and cross sections | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Isotope** | **conc** | **Coh b** | **Inc b** | **Coh xs** | **Inc xs** | **Scatt xs** | **Abs xs** |
| H | --- | -3.7390 | --- | 1.7568 | 80.26 | 82.02 | 0.3326 |
| 1H | 99.985 | -3.7406 | 25.274 | 1.7583 | 80.27 | 82.03 | 0.3326 |
| 2H | 0.015 | 6.671 | 4.04 | 5.592 | 2.05 | 7.64 | 0.000519 |
| 3H | (12.32 a) | 4.792 | -1.04 | 2.89 | 0.14 | 3.03 | 0 |

# Spin Incoherence

For the isotopes H1, H2, and H3, $b_{coh}$ and $b_{inc}$ depend on their:

- $b_-$ and $b_+$ which are the **scattering lengths,**
- $p_-$ and $p^+$ which are the **occupation probabilities**
- for the combined neutron-plus-nucleus system with spins $I - \frac{1}{2}$ or $I + \frac{1}{2}$

For example, H1 has:

- a spin of $I = \frac{1}{2}$
- combined spin with the neutron of 0 and 1 with degeneracies of 1 and 3.

The measured scattering lengths are $b_{-,H1} = -47.5$ and $b_{+,H1} = 10.85$

Therefore, the coherent and incoherent scattering lengths will be

$$b_{\mathrm{coh,\,H1}} = \frac{1}{4}(-47.5) + \frac{3}{4}10.85 = -3.7406$$

$$b_{\mathrm{inc,\,H1}} = \left(\frac{1}{4}(-47.5)^2 + \frac{3}{4}(10.85)^2 - b_{\mathrm{coh,\,H1}}^2\right)^{\frac{1}{2}} = 25.272$$

| Neutron scattering lengths | | | | |
|---|---|---|---|---|
| **Isotope** | **conc** | **Coh b** | **Inc b** | **Coh** |
| H | --- | -3.7390 | --- | 1.75 |
| 1H | 99.985 | -3.7406 | 25.274 | 1.75 |
| 2H | 0.015 | 6.671 | 4.04 | 5.59 |
| 3H | (12.32 a) | 4.792 | -1.04 | 2.89 |

# Isotopic incoherence

Argon isotopes are all zero spin nuclei which lead to zero $b_{inc}$ values. Unlike for the hydrogen atoms, there will be no contribution to the incoherent scattering from **spin incoherence**.

The $b_{inc}$ for the Ar atom type arises from **isotopic incoherence**.

The natural abundances of the argon isotopes are 0.337%, 0.063%, and 99.6% for the Ar36, Ar38, and Ar40 isotopes, respectively.

The coherent and incoherent scattering lengths of argon are:

| Atom Type | $b_{coh}$ / fm | $b_{inc}$ / fm |
|---|---|---|
| Ar36 | 24.9 | 0.0 |
| Ar38 | 3.5 | 0.0 |
| Ar40 | 1.83 | 0.0 |
| Ar | 1.909 | 1.3381 |

$$b_{\mathrm{coh,\,Ar}} = 0.00337 \times 24.9 + 0.00063 \times 3.5 + 0.996 \times 1.83 = 1.909$$

$$b_{\mathrm{inc,\,Ar}} = (0.00337 \times (24.9)^2 + 0.00063 \times (3.5)^2 + 0.996 \times (1.83)^2 - b_{\mathrm{coh,\,Ar}}^2)^{\frac{1}{2}} = 1.3381$$

# Isotopic incoherence

When we use the Ar atom type in MDANSE, we assume that the isotopes of argon in our system are randomly distributed and follow their natural abundances.

In the case of the atom types which are not specific isotopes (e.g. H and Ar) the $b_{inc}$ contains contributions from only spin and isotopic incoherence.

| Atom Type | $b_{coh}$ / fm | $b_{inc}$ / fm |
|---|---|---|
| Ar36 | 24.9 | 0.0 |
| Ar38 | 3.5 | 0.0 |
| Ar40 | 1.83 | 0.0 |
| Ar | 1.909 | 1.3381 |

# Chemical incoherence

Incoherent scattering can arise from a system where two or more atom types with **different scattering lengths** share **similar trajectories and distributions** across the system.

Incoherent scattering from this source of randomness is known as ***chemical incoherence***.

Chemical incoherence is included in MDANSE but will turn up in the coherent signal, (dynamic coherent structure factor) rather than the incoherent (dynamic incoherent structure factor).

# Scenario of the tutorial

We will create three new atom types, `Atm1`, `Atm2`, and `Atm1Atm2`.

`Atm1Atm2` will be a combined atom type, representing a mixture of Atm1 and Atm2.

We will calculate the dynamic coherent structure factor (**DCSF**) and dynamic incoherent structure factor (**DISF**) for:

- a 50/50 mixture of `Atm1` and `Atm2`
- a system of 100% `Atm1Atm2`.

We will then combine the DCSF and DISF results using the neutron total dynamic structure factor (NTDSF) job.

We will compare the coherent and incoherent signal between the two systems.

# The trajectories

We have already generated a trajectory of argon in another exercise.

We are going to re-use it and change the chemical elements assigned to atoms.

NOTE:

You may want to start the MDANSE GUI in the directory of this tutorial, using

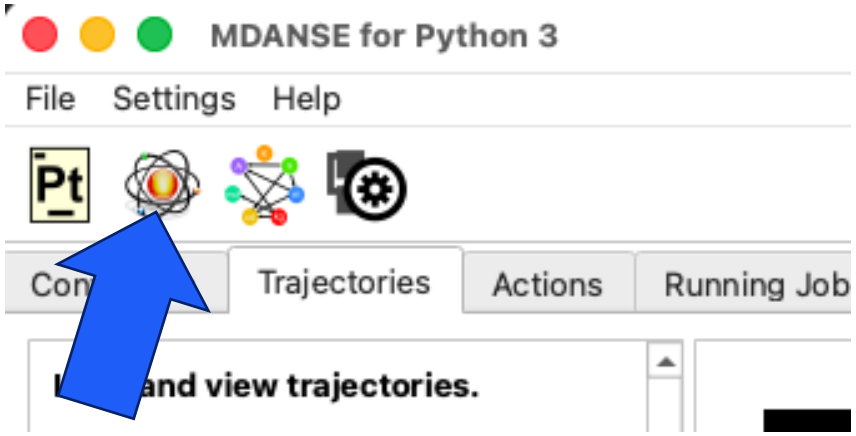
```
mdanse_gui --local
```

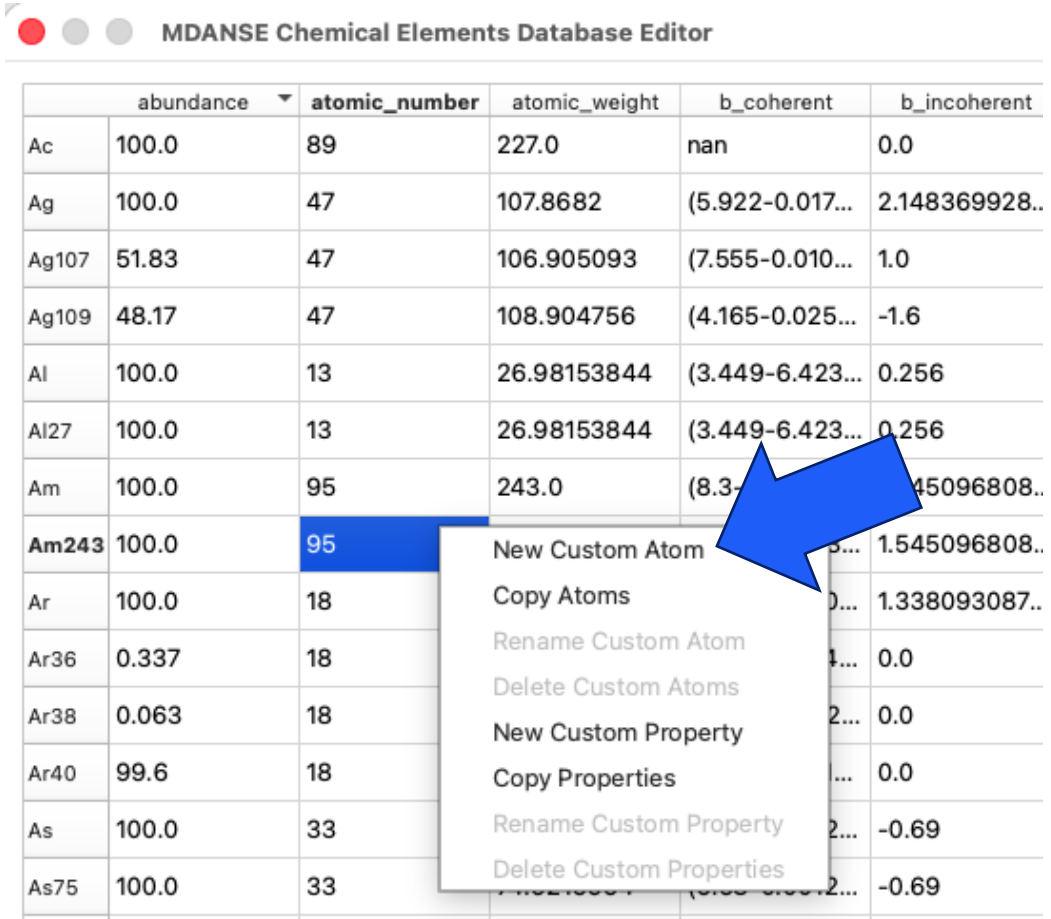
This will keep the changes to the atom database within the working directory, not affecting the central settings in the home directory.

At the same time, you will **not** have access to the settings and definitions you already created in the previous tutorials.

# Adding new chemical elements

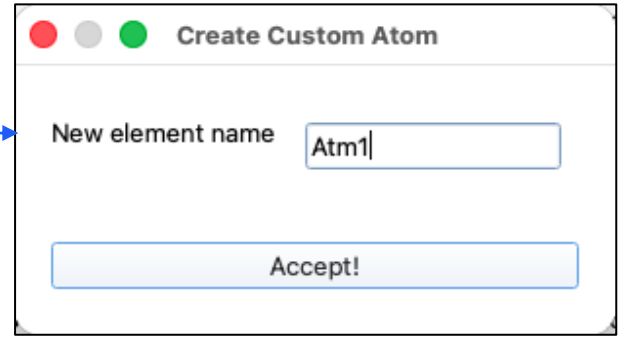
- In the MDANSE GUI, open the chemical elements editor.
- Right-click the table to open the context menu.
- Choose “New Custom Atom”.



# Adding new chemical elements

- Create new atoms called “Atm1”, ”Atm2” and “Atm1Atm2”.
- Set their scattering lengths to the values in the table.
- Assign different colours to the elements.
- Set their van der Waals radius to 0.188 to make them visible in the 3D view. Or give them **different radii** to tell them apart?
- You should expect the new entries to look like this:




| Atom Type | $b_{coh}$ / fm | $b_{inc}$ / fm |
|---|---|---|
| Atm1 | 1.0 | 0.0 |
| Atm2 | 10.0 | 0.0 |
| Atm1Atm2 | 5.5 | 4.5 |

MDANSE Chemical Elements Database Editor

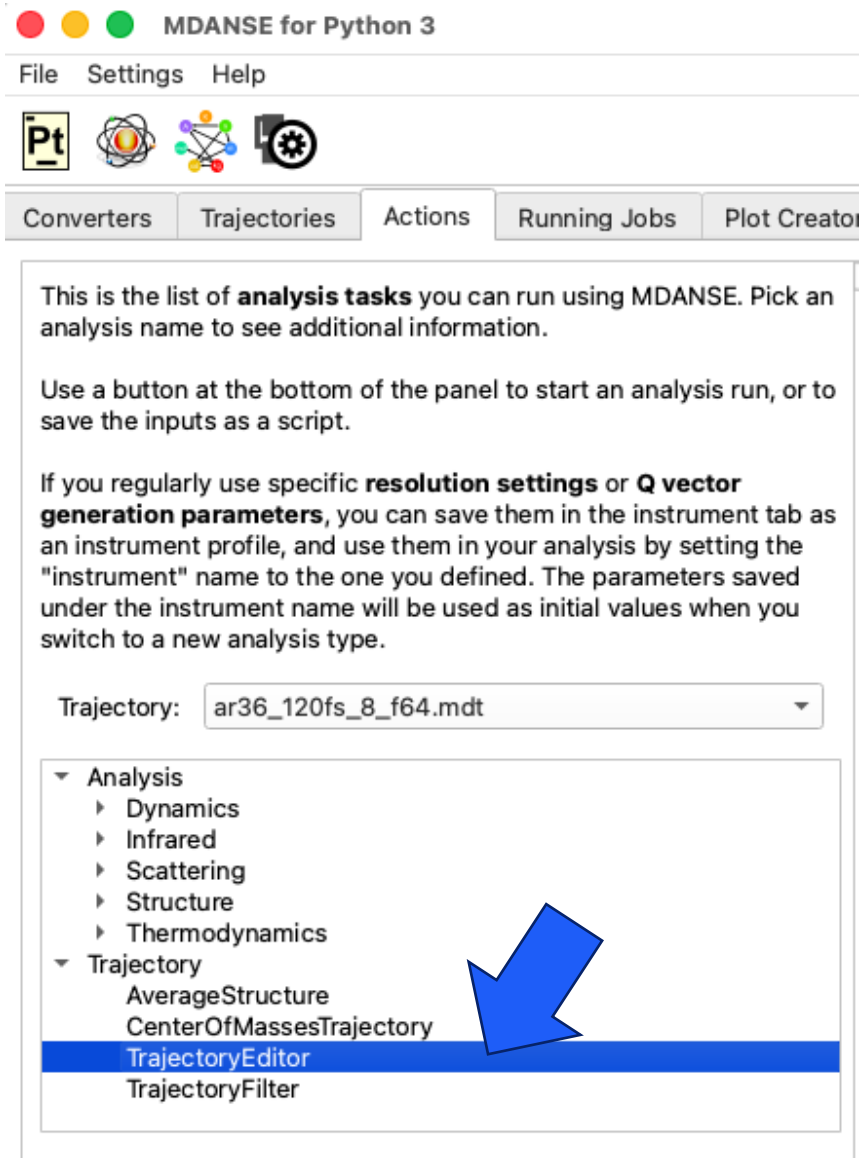
| | abundance | atomic_number | atomic_weight | b_coherent | b_incoherent | block | color |
|---|---|---|---|---|---|---|---|
| Atm1 | 0.0 | 0 | 0.0 | 1.0 | 0.0 | | 255;24;0 |
| Atm2 | 0.0 | 0 | 0.0 | 10.0 | 0.0 | | 207;0;255 |
| Atm1Atm2 | 0.0 | 0 | 0.0 | 5.5 | 4.5 | | 0;54;255 |

# Replacing atoms in the trajectory

- Load the argon trajectory and select TrajectoryEditor.
- The input field you will need is Atom transmutation.




- To replace all atoms with “Atm1Atm2, open the Atom transmutation helper.



# Replacing atoms in the trajectory

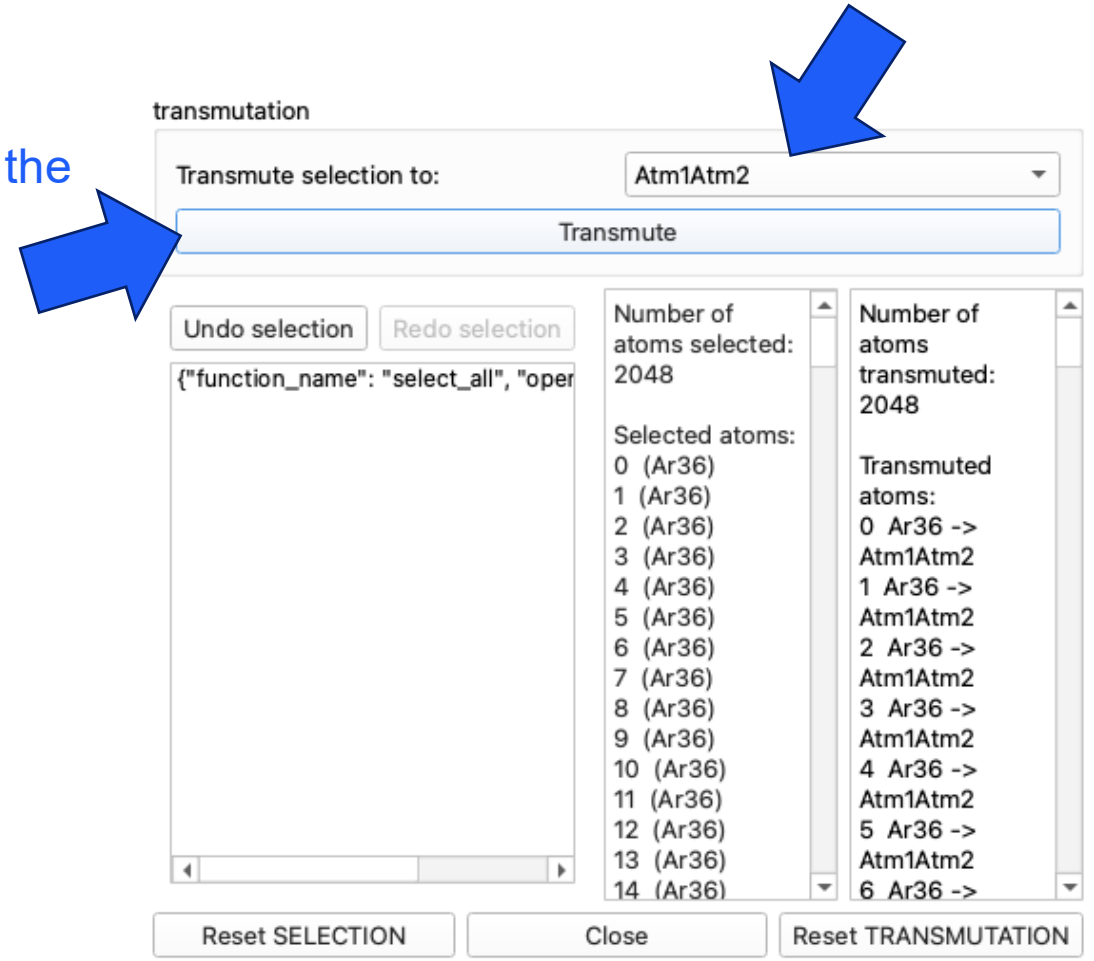
To replace all atoms with “Atm1Atm2”, open the Atom transmutation helper.

Since all the atoms are selected by default, you need to set the atom type to Atm1Atm2, then click “Transmute”.

The bottom-right text field should now contain the list of transmuted atoms.

All that’s left to do is:

- close the dialog,
- change the output file name,
- click “run”.



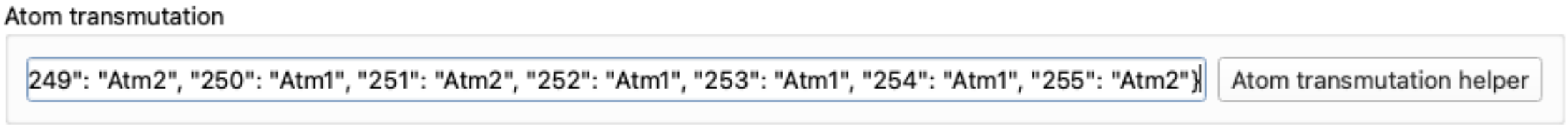
# Replacing atoms in the trajectory

In the second trajectory, we want assign Atm1 and Atm2 types to 50% of the atom each, randomly.

There is no option of doing it automatically in the GUI, so instead we will run the script “`mdanse_inputs/script2_random.py`”

The script will output a text string that can be pasted directly into the Atom transmutation text field, without even opening the helper dialog.

In the end, that part of the GUI will look similar to this:




Again, you can now change the output file name and click “run”.

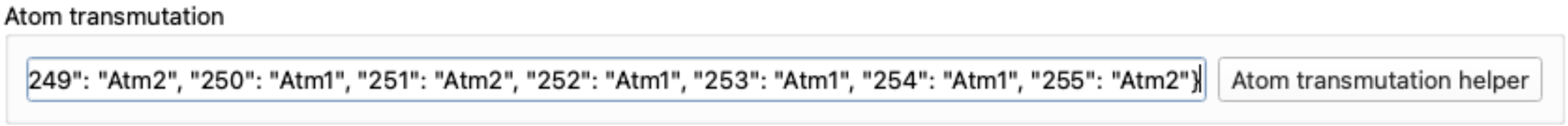
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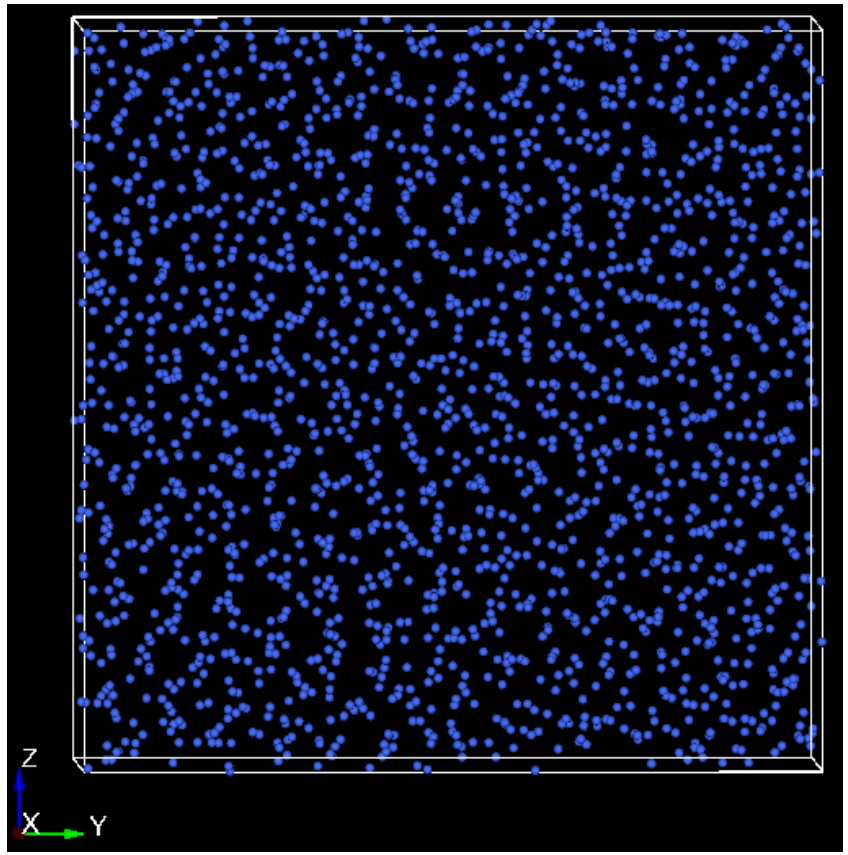
In the end, that part of the GUI will look similar to this:

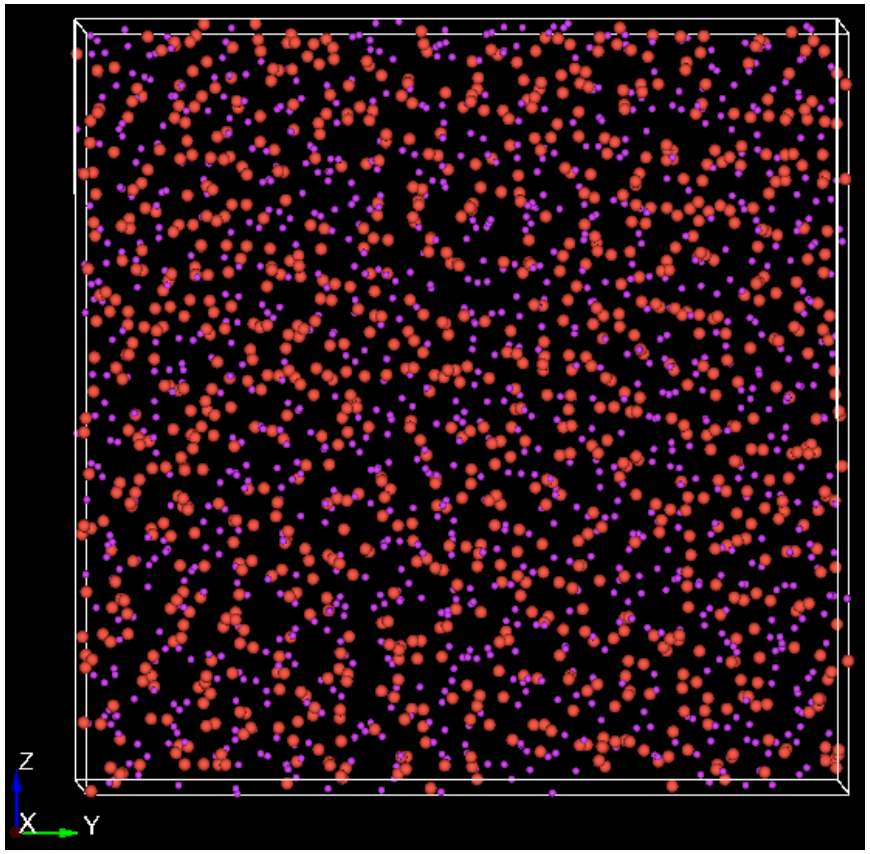



Again, you can now change the output file name and click “run”.

# Replacing atoms in the trajectory

Now, you should have two trajectories, looking like this:



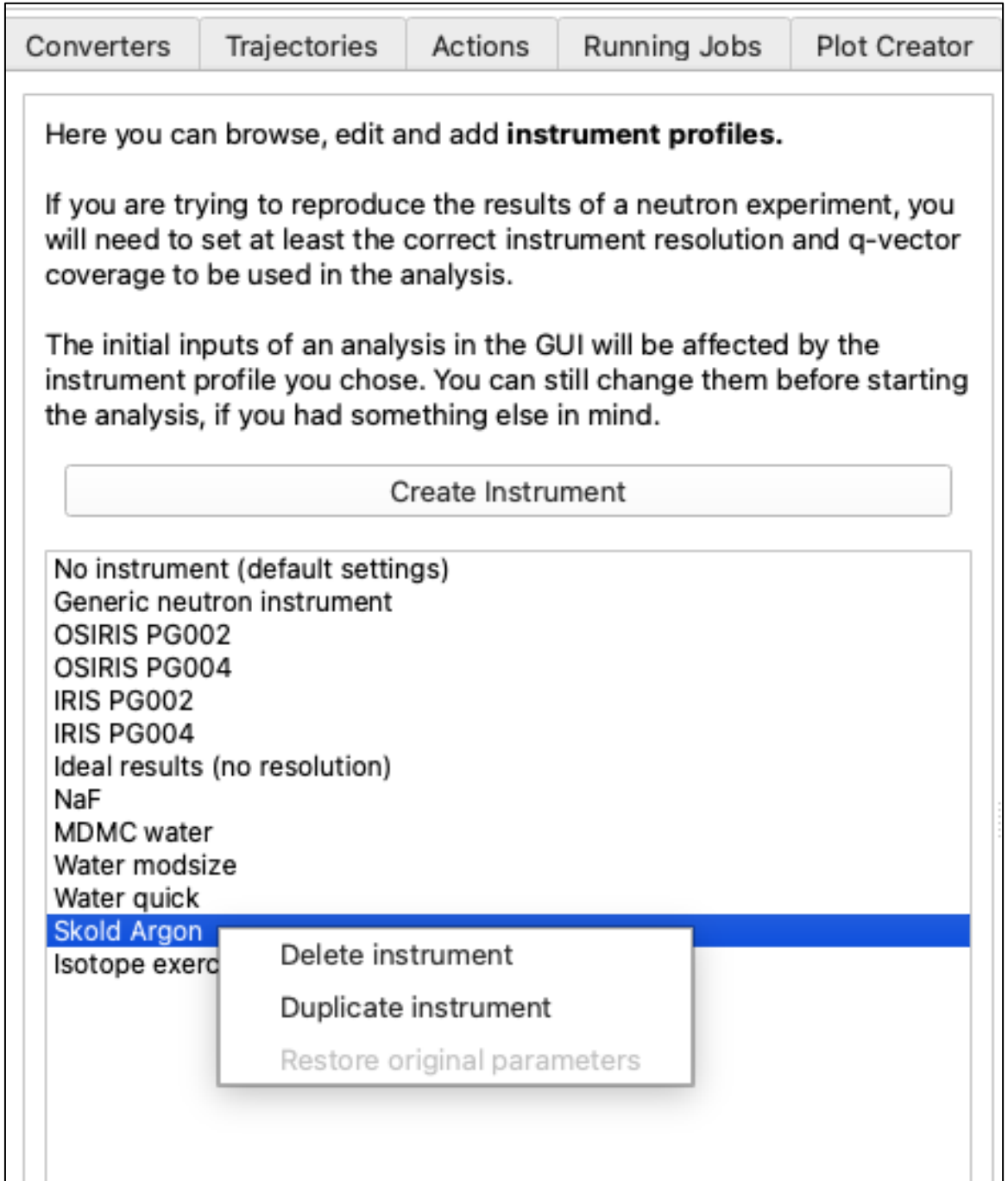
# DCSF, DISF and NDTSF calculations

We can start calculating the neutron observables for both trajectories.

In the **Instruments tab**, create a new instrument.

(Do you still have the instrument profile you created in the previous tutorial? You can use it as the starting point.

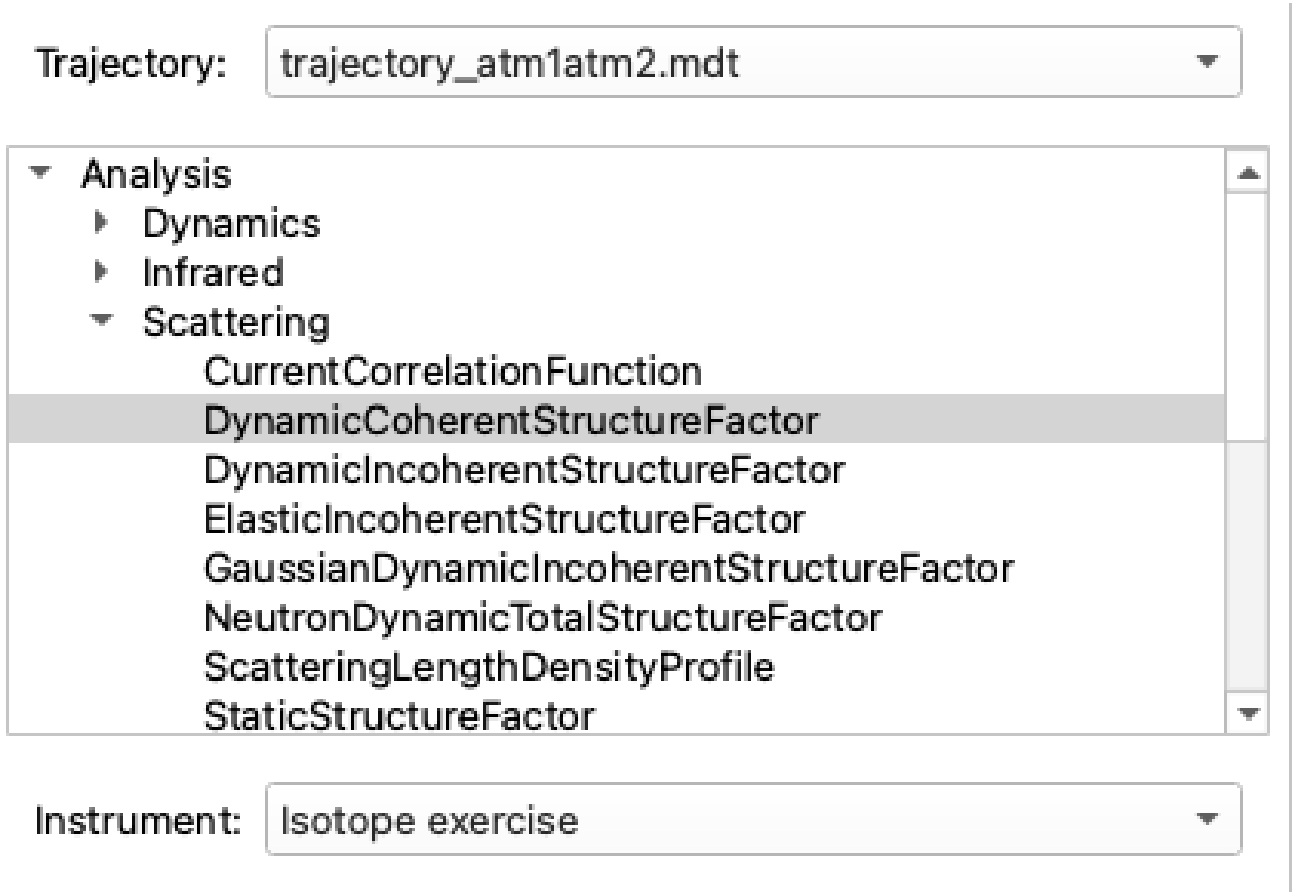
In the instrument tab, right-click the previous instrument definition you created and click “Duplicate instrument”. Then edit the new instrument.)

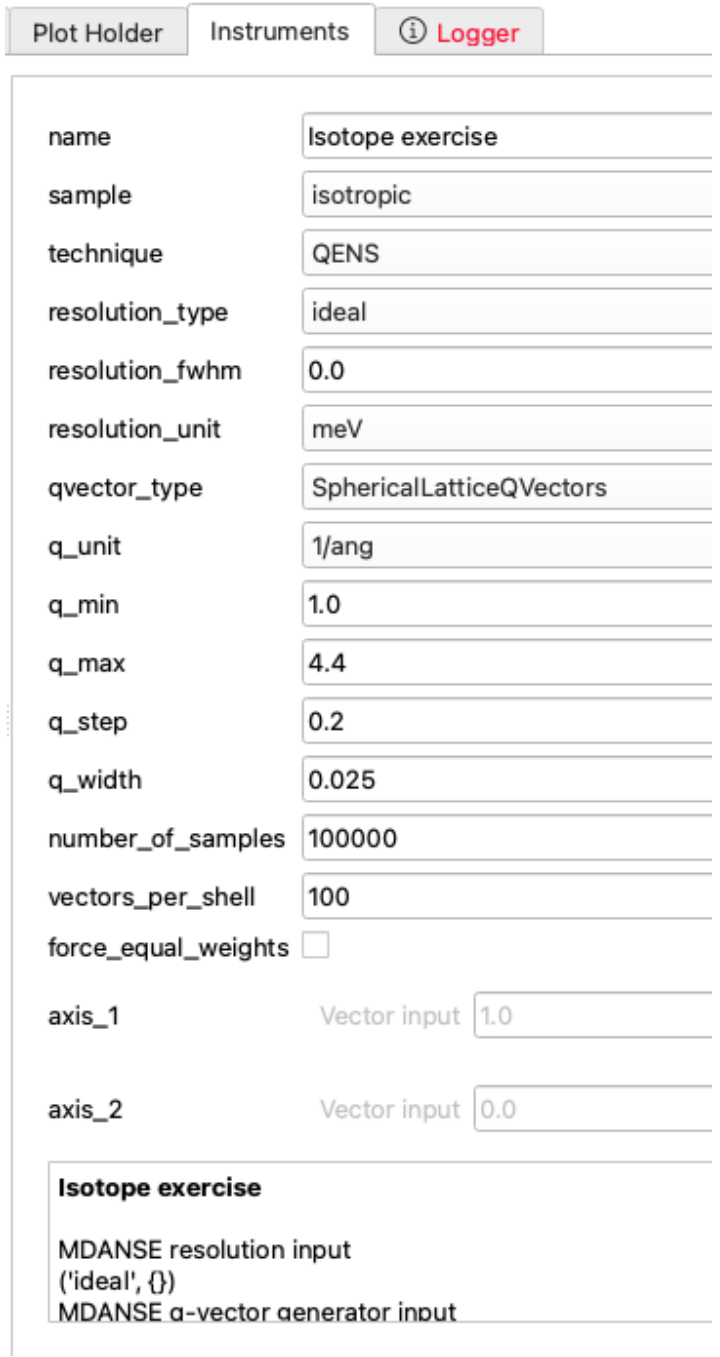


# DCSF, DISF and NDTSF calculations

Set the instrument parameters as shown here:

In the Actions tab, select the new instrument and the first calculation type: DynamicCoherentStructureFactor.



# DCSF, DISF and NDTSF calculations

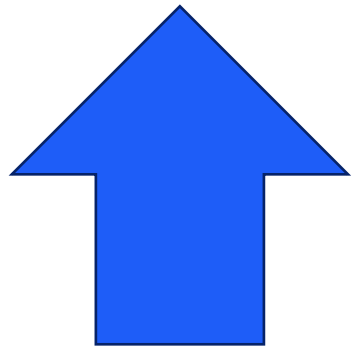
In the Actions tab, select the new instrument and the first calculation type: DynamicCoherentStructureFactor.

The Q-vector generator parameters should be the same as here:

Q vector generator

Generator type: SphericalLatticeQVectors

| | | | |
|---|---|---|---|
| 1 | seed | 0 | IntegerConfig... |
| 2 | shells | [10.0,44.0,2.0] | RangeConfig... |
| 3 | n_samples | 100000 | IntegerConfig... |
| 4 | n_vectors | 100 | IntegerConfig... |
| 5 | force_equal_... | False | BooleanConfi... |
| 6 | width | 0.25 | FloatConfigur... |



Trajectory: trajectory_atm1atm2.mdt

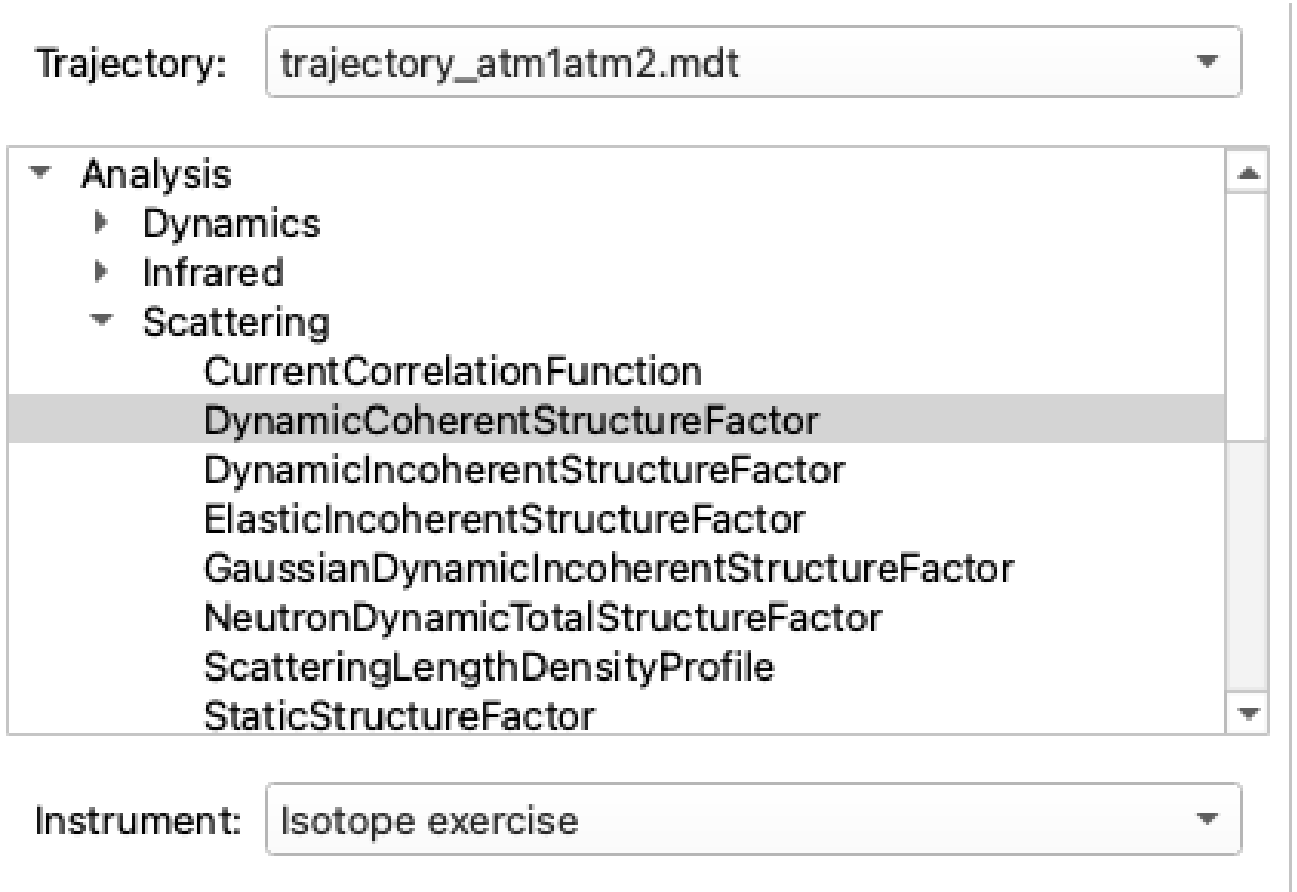
- Analysis
  - Dynamics
  - Infrared
  - Scattering
    - CurrentCorrelationFunction
    - DynamicCoherentStructureFactor
    - DynamicIncoherentStructureFactor
    - ElasticIncoherentStructureFactor
    - GaussianDynamicIncoherentStructureFactor
    - NeutronDynamicTotalStructureFactor
    - ScatteringLengthDensityProfile
    - StaticStructureFactor

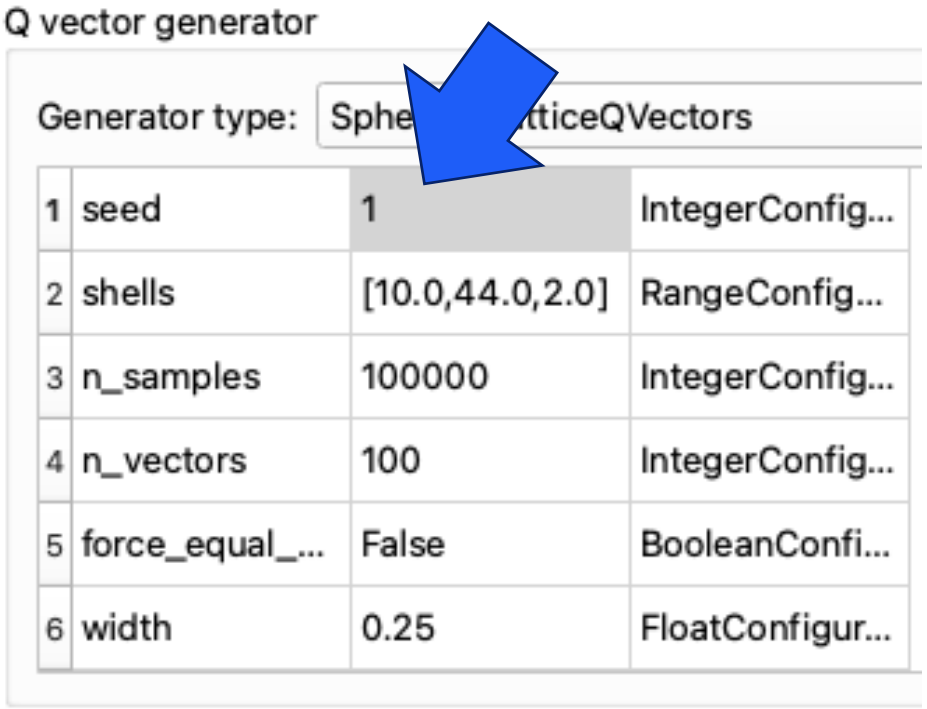
Instrument: Isotope exercise

# DCSF, DISF and NDTSF calculations

In the Actions tab, select the new instrument and the first calculation type: DynamicCoherentStructureFactor.

The Q-vector generator parameters should be the same as here:



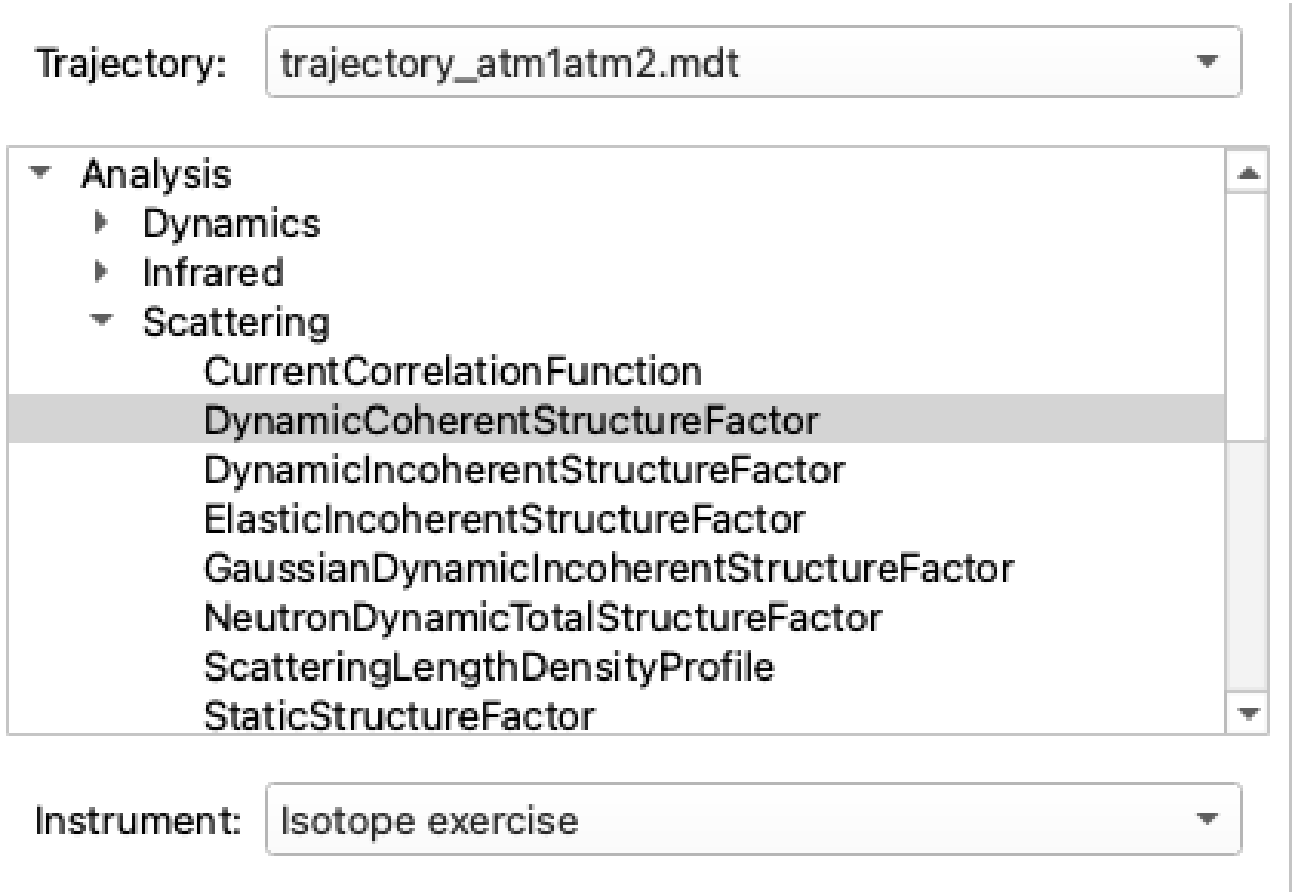



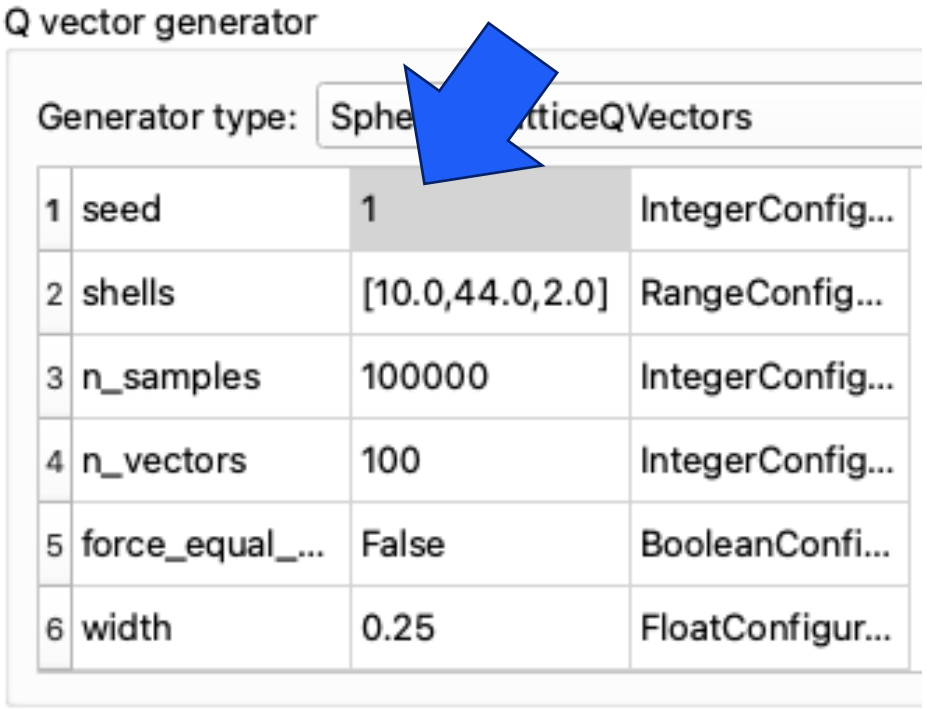

NOTE:
We set the random seed to 1 to make sure that the results are reproducible between repeated calculations.

# DCSF, DISF and NDTSF calculations

In the Actions tab, select the new instrument and the first calculation type: DynamicCoherentStructureFactor.

The Q-vector generator parameters should be the same as here:



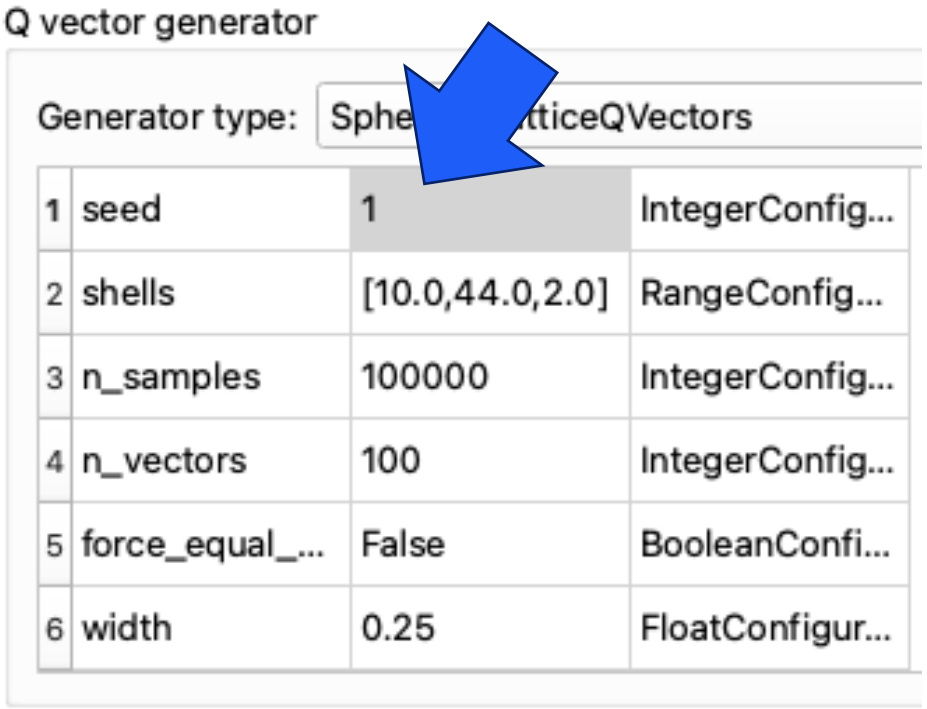

NOTE:
We set the random seed to 1 to make sure that the results are reproducible between repeated calculations.

# DCSF, DISF and NDTSF calculations

Run the **DCSF** and **DISF** calculations, for **both trajectories**, while giving the output files meaningful names, such as:

- mdanse_outputs/disf_atm1atm2.mda
- mdanse_outputs/dcsf_atm1atm2.mda

- mdanse_outputs/disf_atm1_atm2.mda
- mdanse_outputs/dcsf_atm1_atm2.mda




NOTE:
We set the random seed to 1 to make sure that the results are reproducible between repeated calculations.

# DCSF, DISF and NDTSF calculations

You can check on the progress of your analysis in the “Running Jobs” tab.

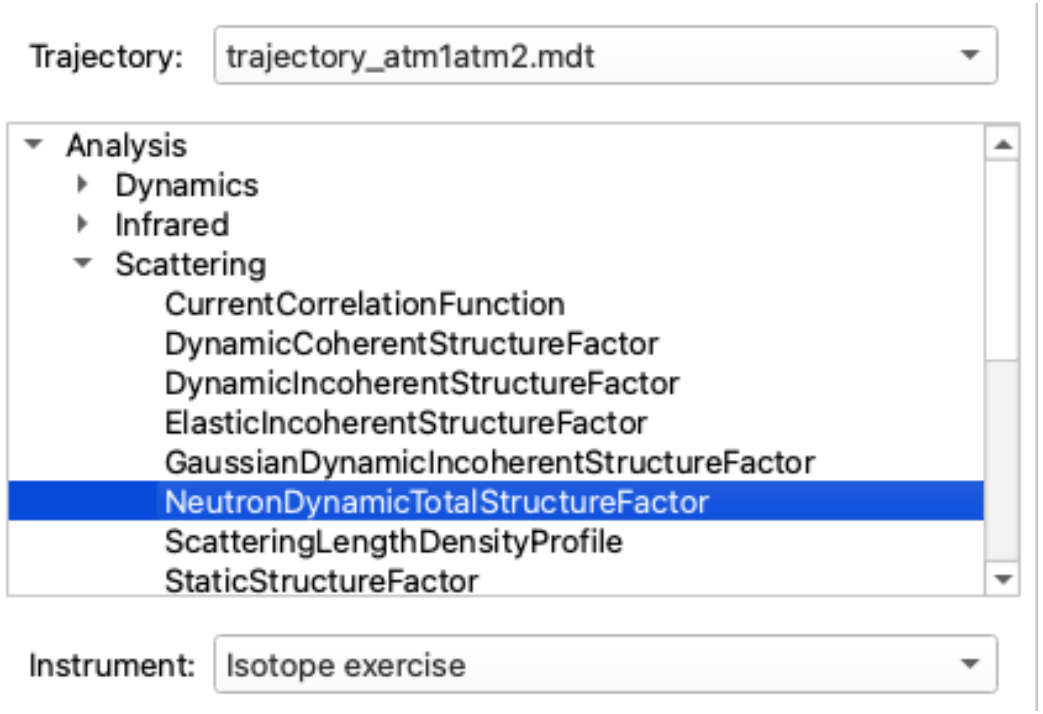
It is fine to run multiple analysis tasks at the same time, as long as you have enough CPU cores and memory to spare.



# DCSF, DISF and NDTSF calculations

Run the NeutronDynamicTotalStructureFactor.

Match the trajectory name to the input file names.




- mdanse_outputs/disf_atm1atm2.mda
- mdanse_outputs/dcsf_atm1atm2.mda

MDANSE Coherent Structure Factor

dcsf.mda

MDANSE Incoherent Structure Factor

disf.mda

# DCSF, DISF and NDTSF calculations

Run the NeutronDynamicTotalStructureFactor.

Match the trajectory name to the input file names.

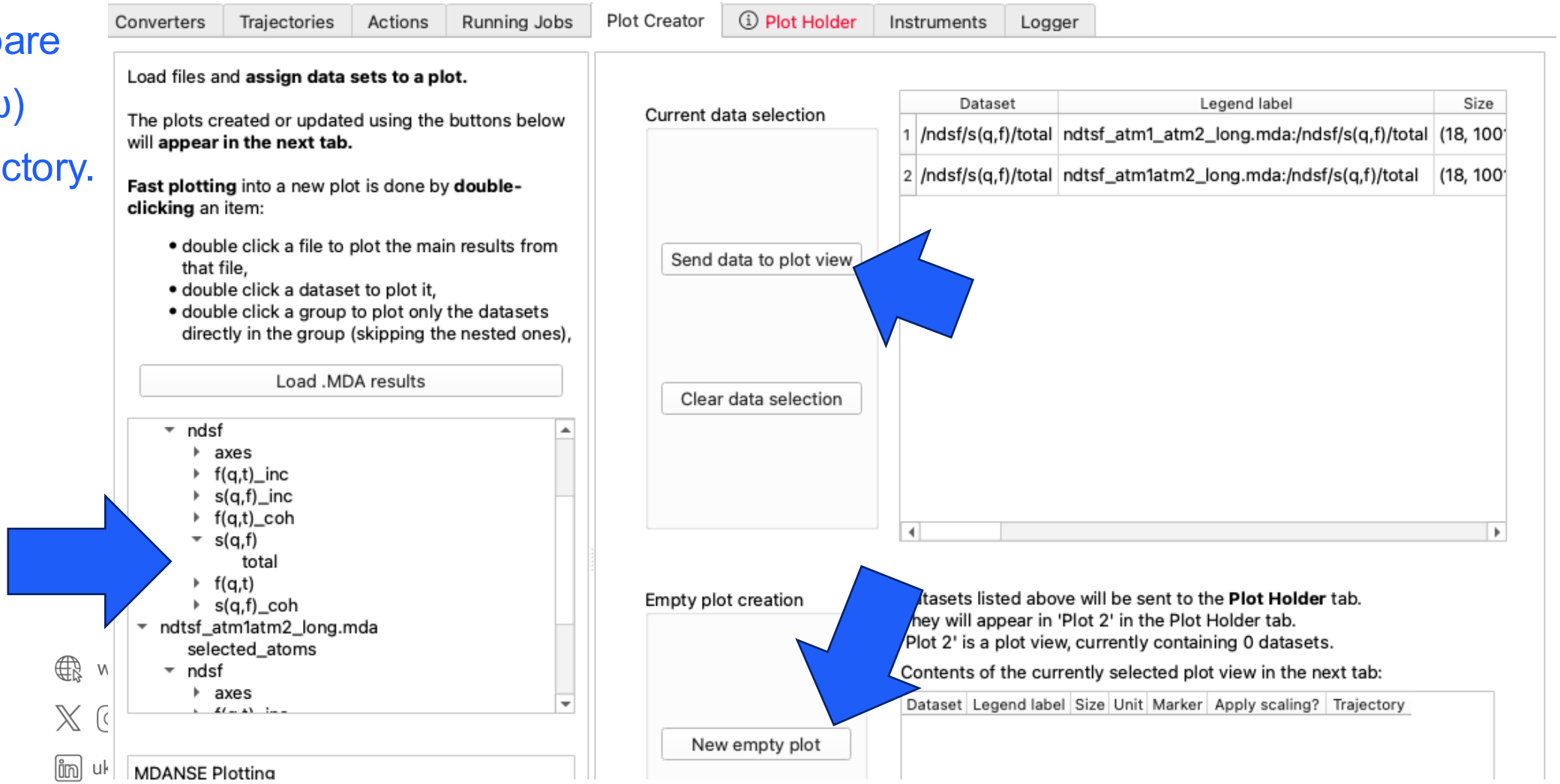
Here, we assume that you used output file names:

- mdanse_outputs/ndtsf_atm1atm2.mda
- mdanse_outputs/ndtsf_atm1_atm2.mda

# DCSF, DISF and NDTSF calculations

In the Plot Creator tab, create plots to compare the results.

Here, we compare the total S(Q, ω) from each trajectory.

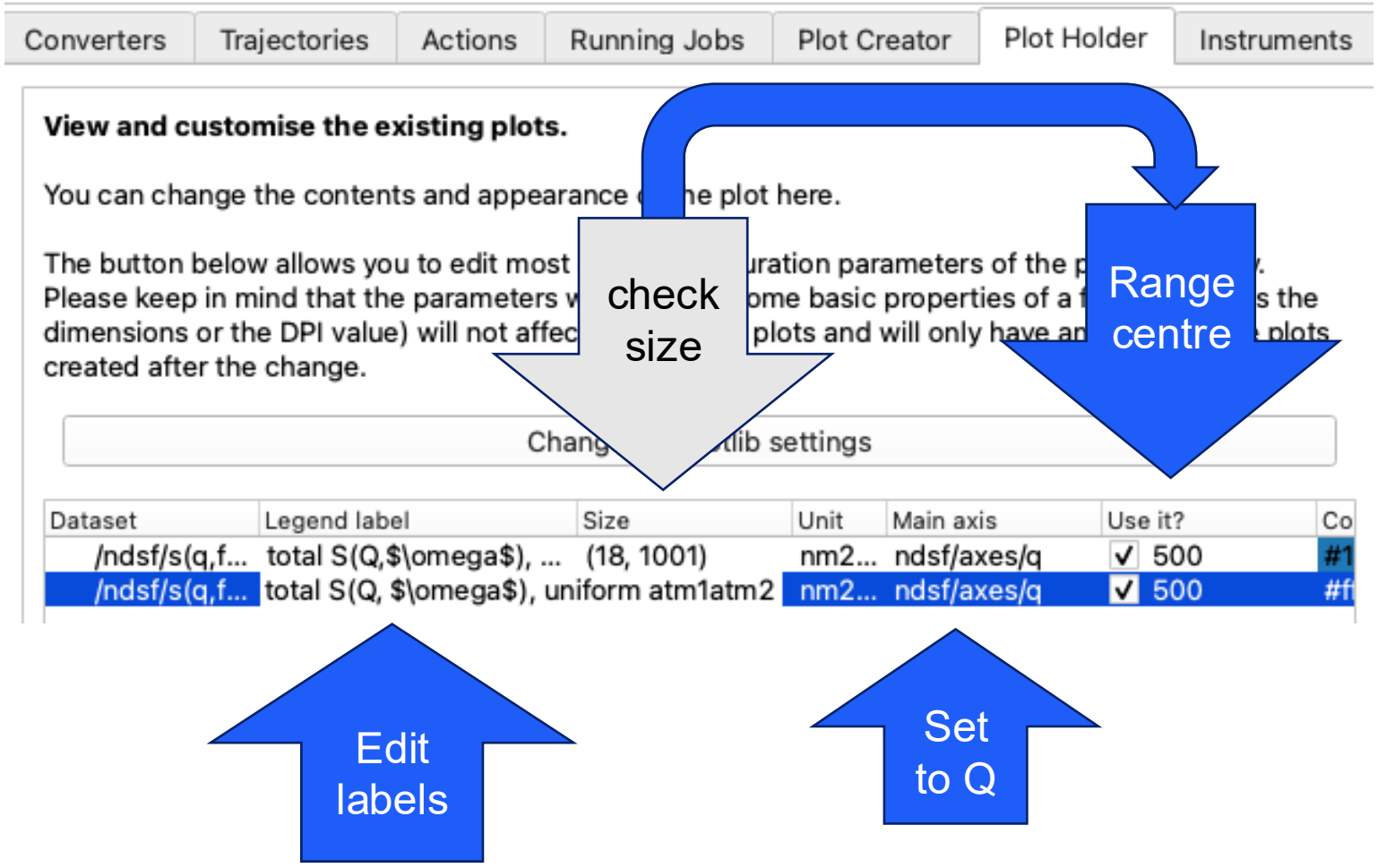


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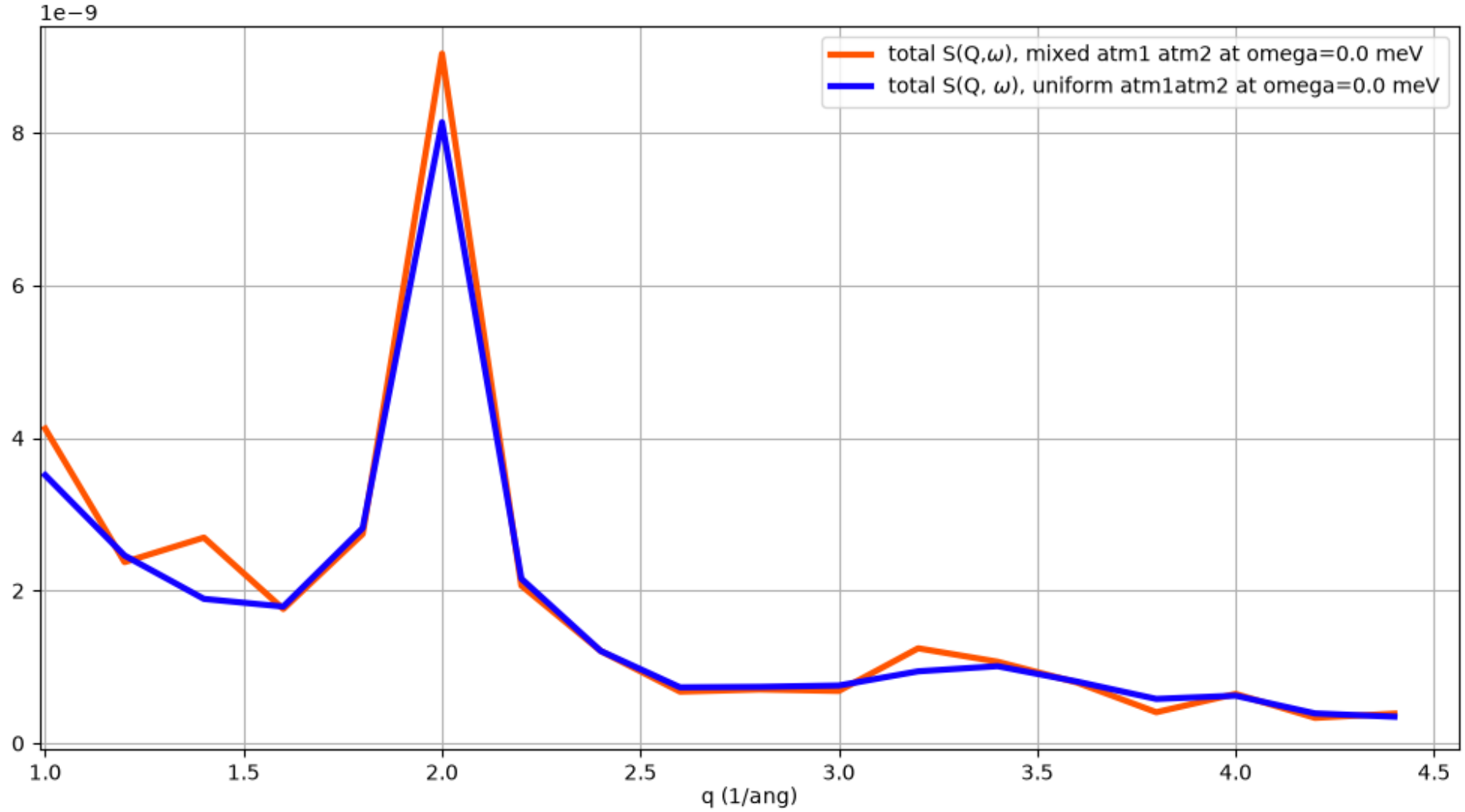
Plot the results as a function of Q,
and pick the data row in the middle of
the range (which is E=0).



# DCSF, DISF and NDTSF calculations

In the Plot Creator tab, create plots to compare the results.

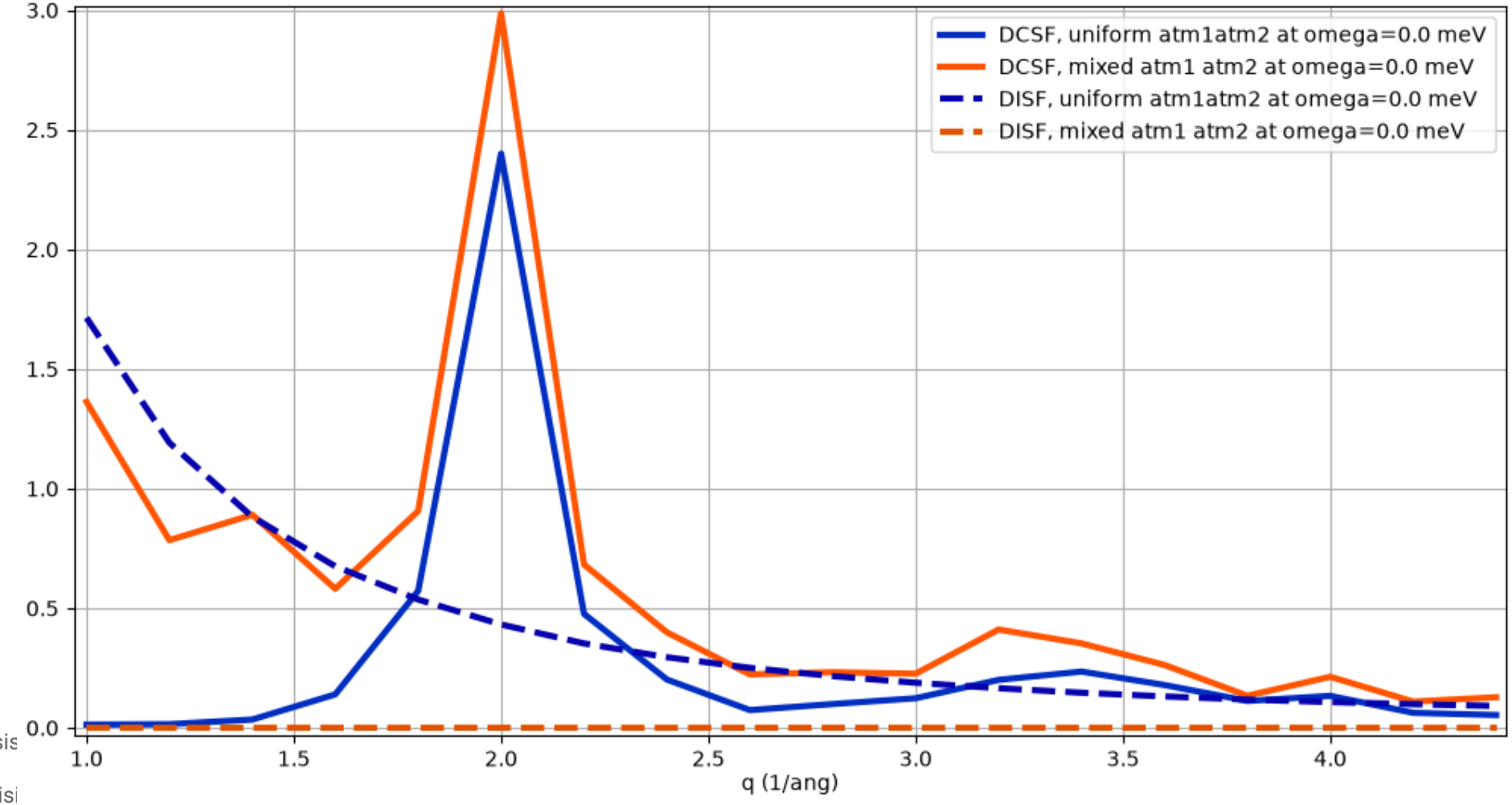
Here, we compare the total S(Q, ω) from each trajectory.



# DCSF, DISF and NDTSF calculations

In the Plot Creator tab, create plots to compare the results.

Here, we compare the coherent and incoherent components of S(Q, ω) from each trajectory.



# Interpretation of the results

A longer trajectory would have produced a less noisy result.

Overall, the total scattering produced by the two trajectories converges to the same result.

There is a difference between a trajectory where every atom is of an **exactly known, specific type**,

and a trajectory which uses the **averaged atom type**.

The trajectory with a mix of Atm1 and Atm2 does not show the isotopic incoherence contribution to the incoherent signal.

